

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Brainstorming & Idea Prioritisation

Online Complaint Registration System — MERN Stack Complaint
Management Platform

Phase: Brainstorming & Ideation Phase

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Brainstorming & Idea Prioritisation

1. Introduction

This document records the structured brainstorming and ideation process undertaken during the inception of the Online Complaint Registration System project [cite: 1136]. The objective of this phase was to identify challenges associated with traditional complaint handling mechanisms, generate candidate solution ideas, evaluate alternative approaches, and systematically prioritise features using established brainstorming and prioritisation frameworks [cite: 1137].

The outcomes of this phase directly inform the Problem Statement Definition, Requirement Analysis, System Design, and Development planning cycles [cite: 1138]. The final project targets a centralized, transparent platform connecting citizens, operating service agents, and governing administrators seamlessly [cite: 1137].

2. Problem Statement Selection

Traditional public and private service complaint registration mechanisms frequently depend on manual procedures, fragmented communication channels, and opaque workflows. Citizens regularly suffer from extreme processing updates visibility gaps, long communication delays with assigned personnel, and the absence of end-to-end status tracking tools.

Concurrently, organizational supervisors struggle with inefficient assignment workflows, zero oversight metrics over agent processing backlogs, and non-aggregated compliance records [cite: 1140]. Therefore, there is a clear academic and practical requirement for a centralized, multi-tenant MERN stack solution providing real-time notification loops, secure authentication, and metrics-driven dashboard transparency [cite: 1140].

3. Idea Generation & Grouping

3.1 Raw Idea Listing

The initial unstructured idea compilation was recorded, filtered, and checked for core structural feasibility against architectural rules [cite: 1141].

Idea ID [cite: 1143]	Idea Description [cite: 1143]	Category [cite: 1143]	Retained? [cite: 1143]
I-01 [cite: 1144]	Build a MERN-based complaint management platform for full architectural customization control. [cite: 1141]	Core Architecture [cite: 1141]	Yes [cite: 1141]
I-02	Use email-based complaint submission only.	Existing Solutions	 Discarded – Limited tracking
I-03	Use spreadsheet-based complaint tracking across departments.	Existing Solutions	 Discarded – Poor scalability
I-04 [cite: 1144]	Implement JWT-based stateless authentication utilizing secure HTTP-only cookies. [cite: 1141]	Security [cite: 1141]	 Yes [cite: 1141]
I-05 [cite: 1144]	Implement Role-Based Access Control distinguishing Citizen, Agent, and Admin views. [cite: 1141]	Access Control [cite: 1141]	 Yes [cite: 1141]
I-06	Develop an interactive Citizen Dashboard for ticket logging and monitoring.	User Management	Yes
I-07	Develop an operational Agent Workspace dashboard for tracking assigned cases.	Operations	Yes
I-08 [cite: 1144]	Build an interactive Admin Dashboard with real-time resolution metrics. [cite: 1141]	Administration [cite: 1141]	Yes [cite: 1141]
I-09 [cite: 1144]	Automate complaint assignment and tracking dispatch workflows natively.	Complaint Management	Yes
I-10	Integrate a real-time instant chat system between citizens and agents using Socket.IO.	Communication	Yes
I-11	Real-time push notification system for status lifecycle modifications.	User Experience	Yes
I-12	Integrate post-resolution citizen feedback, star ratings, and textual reviews.	Quality Assurance	Yes
I-13	Analytics engine for operational overview, performance metrics, and reporting insights.	Analytics	Yes
I-14 [cite: 1144]	Implement AI-driven automated complaint categorization and priority routing features. [cite: 1151]	Advanced Features [cite: 1156]	 Future Scope [cite: 1157]

Idea ID [cite: 1143]	Idea Description [cite: 1143]	Category [cite: 1143]	Retained? [cite: 1143]
I-15 [cite: 1158]	Develop mobile native application wraps across platforms. [cite: 1159]	Marketplace Features [cite: 1160]	 Future Scope [cite: 1157]

3.2 Idea Grouping – Affinity Diagram

To establish clean conceptual boundaries, the generated ideas are categorized into structural domain branches below [cite: 1161].

FIGURE 1.1 – AFFINITY / MIND-MAP: IDEA GROUPING BY DOMAIN [CITE: 1179]

4. Idea Prioritisation – MoSCoW Framework

The MoSCoW framework maps system components out to establish a transparent, strategic deployment pathway for continuous implementation sprints [cite: 1180, 1181].

Priority [cite: 1182]	Classification [cite: 1182]	Feature / Idea [cite: 1182]	Rationale [cite: 1182]
Must Have [cite: 1182]	Core — Release 1 [cite: 1182]	User Registration & Login [cite: 1182]	Foundational constraint to securely isolate consumer data profiles [cite: 1182].
Must Have [cite: 1182]	Core — Release 1 [cite: 1182]	JWT Authentication [cite: 1182]	Ensures secure, stateless, cookie-bound authorization mapping [cite: 1182].
Must Have	Core — Release 1	Citizen Dashboard	Primary tracking landing module for problem submissions.
Must Have	Core — Release 1	Agent Dashboard [cite: 1182]	Operational node required to monitor and update case lifecycles [cite: 1182].
Must Have	Core — Release 1	Admin Dashboard	Central administration cockpit ensuring supervisory capability.
Must Have [cite: 1182]	Core — Release 1 [cite: 1182]	Complaint Registration [cite: 1182]	Core business logic function [cite: 1182].
Must Have [cite: 1182]	Core — Release 1 [cite: 1182]	Complaint Tracking [cite: 1182]	Essential flow parameter giving visibility to citizens [cite: 1182].
Must Have	Core — Release 1	Complaint Assignment Workflow	Enables supervisors to assign items to active handling agents.
Must Have	Core — Release 1	Real-Time Chat System	Socket-driven module to link citizens and agents directly.
Must Have	Core — Release 1	Notification System	Drives automated triggers on lifecycle milestone changes.
Must Have	Core — Release 1	Feedback System	Measures final user satisfaction and service quality standards.
Should Have [cite: 1182]	Release 1 [cite: 1182]	Analytics Dashboard [cite: 1182]	Enables data-driven business decision-making and performance monitoring [cite: 1182].
Should Have	Release 1	Agent Approval Workflow	Restricts network workspace access to explicitly approved handlers.
Should Have [cite: 1182]	Release 1 [cite: 1182]	MongoDB Atlas Integration [cite: 1182]	Provides resilient, cloud-managed cluster horizontal scaling [cite: 1182].

Priority [cite: 1182]	Classification [cite: 1182]	Feature / Idea [cite: 1182]	Rationale [cite: 1182]
Could Have [cite: 1183]	Release 2 [cite: 1183]	Mobile Application [cite: 1183]	Improves platform availability but is not a blocking dependency to initial release [cite: 1183].
Could Have [cite: 1183]	Release 2 [cite: 1183]	AI Complaint Categorization [cite: 1183]	Non-trivial automated parsing deferred to future roadmaps [cite: 1183].
Could Have	Release 2	Multi-language Support	Expands digital inclusion boundaries for diverse user groups.
Could Have	Release 2	SMS Notifications	Secondary fallback alert route channel.

5. Conclusion

The brainstorming and ideation phase successfully transformed the project vision into a structured set of prioritized deliverables and core implementation objectives [cite: 1185]. The selected Must Have features define the complete Minimum Viable Product (MVP) scope for Sprint 1 and Sprint 2 execution [cite: 1186]. All accepted operational ideas map directly to formal User Stories defined inside the subsequent Requirement Analysis documentation layers [cite: 1186].